Justifying the use of the VIX, through replication

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1 Introduction

We have made some Important definitions from the VRP to the payoff of a variance swap. I have recently taught myself an introduction to stochastic calculus only learning the brief steps. I have learnt the material from the work of Neuberger, A. (1994). "The Log Contract.", Demeterfi, K., Derman, E., Kamal, M., Zou, J. (1999). "More Than You Ever Wanted to Know About Volatility Swaps.",and Breeden, D. T., Litzenberger, R. H. (1978) and I am continuing to learn more about volatility trading. "Prices of State-Contingent Claims Implicit in Option Prices." I go through each step of the derivation, in great detail as it has really helped me learn and connect different ideas. Firstly, I derived the replication for a variance swaps as done in the seminal work listed above

2 Replicating a variance swap

as mentioned in the previous section, the payoff of variance swap is

$$P(T) = N \times (v(T) - K_{Var})$$

For the no arbitrage condition to hold at time $t = 0$ the payoff must be $P(t) = 0$

$$P(T) = 0 \rightarrow v(T) = K_{Var}$$

$$v(T) = K_{Var} \Leftrightarrow \sigma_R^2 = K_{Var}$$

Where σ_R^2 is the realised variance over the time length of the swap. This lead us to

$$\sigma_R^2 = \frac{1}{T}\int_0^T \sigma_t^2 dt$$

This is the payoff we are trying to replicate. In my other project on my Github Estimating Implied Probabilities in Option Prices. I walk through the derivation of gaining this idea of 'calls as a spanning set', as well as gaining butterfly option prices in terms of a conditional probability density function. From the fundamental theorem of asset pricing, current prices are discounted expected payoffs, for a derivative with value V_t at time t with payoff function $g(S_T)$ at Time T:

$$V_t = e^{-rt} E_t^Q[g(S_T)]$$

This is the expected value of the payoff $g(S_T)$ at Time T under the risk neutral measure, with respect to the discount factor e^{-rt} being the numerarie. I learnt about this in Blyth's "An Introduction to Quantitative Finance", this is also what is the base for my aforementioned previous project. This book taught me the link between risk neutral density, call prices and replicating payoffs. Here I will apply this framework.

Using nothing more than First year university mathematics we can start by addressing this idea of calls as a spanning set.

for any twice differentiable positive function, $g(x)$ we can look at:

$$\int_0^\infty max(x-K,0)g''(K)dK = [(x-K)g'(K)]_0^x + \int_0^x g'(K)dK$$

We can intergrate by parts and get

$$\int_0^\infty max(x-K,0)g''(K)dK = g(x) - g(0) - xg'(0)$$

if we take risk neutral expectations of both sides, with a payout function of $g(S_T)$

$$E_t^Q\left[\int_0^\infty max(x-K,0)g''(K)dK\right] = e^{rt}\int_0^\infty e^{-rt}C_K(t,Tg''(K)dK$$

This gives us, as you will where $E_t^Q[max(x-K,0)]$ is the risk neutral expected payoff of a call option, where $C_k(K,T)$ is the price of a call option at time t with maturity date T and strike K

$$e^{rt}\int_0^\infty C_k(K,T)g''(K)dK$$

This is the left hand side of the equation now looking at the right we simply, remembering we are taking risk neutral expectation of now $g(S_T)$ Now combining this with the left hand side

$$e^{rt}E_t^Q[g(S_T)] = g(0) + g'(0)S_te^{rt} + e^{rt}\int_0^\infty C_k(K,T)g''(K)dK$$

$$E_t^Q[g(S_T)] = g(0)e^{-rt} + g'(0)S_t + \int_0^\infty C_k(K,T)g''(K)dK$$

This is a pretty important result for derivatives, this is because this allows you to replicate any payoff $g(S_T)$ using a strip of call options. However, of course integrating with upper limit of positive infinity, assumes a continuous function over all strike values.

The question we need to answer now is now what is $g(S_T)$ for a variance swap. After reading the final chapter of Blyth's book I have a brief introduction to stochastic calculus and Ito's lemma. This allow me to find a $g(S_T$ for a variance swap. Our variance swap has a payoff based off σ_R^2, the realised volatility. This is our payoff function because the rest of the variables are not changed and are fixed variables agreed upon at inception. we can define our σ_R^2 to be:

$$\sigma_R^2 = \frac{1}{T}\int_0^T \sigma_t^2 dt$$

where σ_t is the instantaneous volatility at any given time t. The realised variance squares these, which is important to avoid positive and negative volatilities cancelling each other out.

Now we can address the stochastic differential equation to model our price movement, for simplicity we assume no jumps. This is an area I am building up the prerequisites to investigate.

$$\underbrace{\frac{dS_t}{S_t}}_{instantaneous\ return} = \underbrace{rdt}_{drift} + \underbrace{\sigma_t dW_t}_{diffusion}$$

Here W_t is a Wiener process, also known as a Brownian motion. This originally came from a biologist examining pollen particles, and there jittery movement. It is a process with independent normal increments. It has important parameters of: starting time $t = 0$ that $W_0 = 0$, and that the difference in $W_s - W_t \sim N(0, s-t)$, and that $W_s - W_t$ is independent of any previous W_t. These parameters will be useful later for when interacting with Ito's Lemma.

The equation above makes intuitive sense; however, it is obviously very limiting when assuming no jumps. Especially, because short variance swaps have a large tail risk. This means by assuming no large jumps, we underestimate the tail risk. If we set $f(S_t) = ln(S_t)$, we can use Ito's lemma:

$$df = f'(S_t)dS_t + \frac{1}{2}f''(S_t)(dS_t)^2$$

$$(dS_t)^2 = (S_t rdt + S_t\sigma_t dW_t) = (S_t rdt)^2 + S_t^2 r\sigma_t dtdW_t + (S_t\sigma_t dW_t)^2$$

if we look at the the $(dW_t)^2$ term, if we look at the example of a coin gaining one pound for heads, losing a pound for tails. However, if we square our results, we end up with just a drift term. As well as dt goes to zero so will its square. We are left with addressing the $dtdW_t$ this is essentially the same as we have $(dW_t)^2 = dt$ so $dtdW_t^{\frac{3}{2}}$ so it also tends to zero, faster than dt. So we are left with:

$$(dS_t)^2 = S_t^2\sigma_t^2 dt$$

So going back to our Ito Lemmas formula:

$$df = f'(S_t)dS_t + \frac{1}{2}f''(S_t)S_t^2\sigma_t^2 dt$$

with $f'(S_t) = \frac{1}{S_t}$ and $f''(S_t) = -\frac{1}{S_t^2}$ we gain:

$$df = \frac{dS_t}{S_t} - \frac{1}{2}\sigma_t^2 dt$$

solving for σ_t^2 we can get:

$$\sigma_t^2 dt = 2\left(\frac{dS_t}{S_t} - dlnS_t\right)$$

now we have something to relate to our initial formula for the realised volatility.

$$\sigma_t^2 dt = 2\int_0^T \frac{dS_t}{S_t} - dlnS_t dt$$

here we can use some more simple calculus to gain, using log laws we get

$$2\int_0^T \frac{dS_t}{S_t} - 2\int_0^T dlnS_t dt = 2\int_0^T \frac{dS_t}{S_t} - 2ln(\frac{S_t}{S_0})$$

$$\frac{1}{T}\int_0^T \sigma_t^2 dt = \frac{2}{T}\int_0^T \frac{dS_t}{S_t} - \frac{2}{T}ln(\frac{S_t}{S_0})$$

Now we have achieved finding our $g(S_t)$, from this we can return back to our replication formula. However, the first term $\frac{2}{T}\int_0^T \frac{dS_t}{S_t}$ is complete path dependent on how the underlying asset moves over time. We overcome this by using the reasoning outlined by Neuberger.we take the expected value under Q:

$$E_t^Q\left[\frac{2}{T}\int_0^T \frac{dS_t}{S_t}\right] = 0$$

This has an expected value of zero, because S_t is a martingale under Q, so at any given point if we look at the dynamic hedging at any point in time t where the current information we have until this time is $\mathcal{F}_t$:

$$E_t^Q[S_{t+1} - S_t|\mathcal{F}_t] = 0$$

This makes sense as no past trends under Q can give you future information about changes in the price. It also makes an important economic argument of there being no 'free lunch', that you cannot make a profit or a loss by dynamically hedging your position.

$$\frac{1}{T}E_t^Q\left[\int_0^T \sigma_t^2 dt\right] = \frac{2}{T}E_t^Q\left[\frac{S_T - S_0}{S_0} - ln\frac{S_T}{S_0}\right]$$

if we choose a new function $h(S_T) = -ln(\frac{S_T}{S_0})$ we can apply this more easily into our previous expected payoff formula. By simple calculus:

$$h'(S_T) = \frac{1}{S_T}$$

and the second derivative:

$$h''(S_T) = -\frac{1}{S_T^2}$$

This is a very important result here. As we can see where the $\frac{1}{K^2}$ weighting comes from. However, our formula was only defined for positive functions $g(x)$, so we introduce a negative function.

$$E_t^Q[h(S_T)] = h(S^*)e^{S^*} + h'(S^*)S_t + \int_0^{S^*} E_t^Q\left[max[K - x, 0]\right] h''(K)dK + \int_{S^*}^{\infty} C_k(K,T)h''(K)dK$$

Conveniently, $E_t^Q\left[max[K-x,0]\right] = P_K(t,T)$, where $P_K(t,T)$ is the definition of a put option under Q. now plugging in our values:

$$E_t^Q[h(S_T)] = -ln(\frac{S^*}{S_0})e^{rt} - \frac{S_T - S^*}{S_T} + \int_0^{S^*} \frac{P_K(t,T)}{K^2}dK + \int_{S^*}^{\infty} \frac{C_K(t,T)}{K^2}dK$$

letting $S^* = S_0$ we get

$$E_t^Q[h(S_T)] = -\frac{S_T - S_0}{S_T} + \int_0^{S_0} \frac{P_K(t,T)}{K^2}dK + \int_{S_0}^{\infty} \frac{C_K(t,T)}{K^2}dK$$

plugging this back into

$$E_t^Q\left[\int_0^T \sigma_t^2 dt\right] = E_t^Q\left[\frac{S_T - S_0}{S_T} - \frac{S_T - S_0}{S_T} + \int_0^{S_0} \frac{P_K(t,T)}{K^2}dK + \int_{S_0}^{\infty} \frac{C_K(t,T)}{K^2}dK\right]$$

Finally, we have our result for the payoff swap. This work was built originally by Neuberger where he introduced the log contract to show the delta hedged strategy isolates quadratic variation. Then built on in the famous Goldman Sachs "More Than You Ever Wanted to Know About Volatility Swaps."

$$E_t^Q\left[\int_0^T \sigma_t^2 dt\right] = E_t^Q\left[\int_0^{S_0} \frac{P_K(t,T)}{K^2}dK + \int_{S_0}^{\infty} \frac{C_K(t,T)}{K^2}dK\right]$$

3 Connection To The VIX

Due to having a lack of historical options data, I had to come up with a work around for this. However, I noticed that the VIX formula was similar, I then read the VIX white paper and realised that they were the same, with slight differences. To start, the VIX method truncates the strikes, when the replication of course assumes $K \to \infty$, there are also differences the VIX uses a time weighted linear interpolation and a few more.

$$VIX = 100 \times \sqrt{\frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2} e^{rT} Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2}$$

where $Q(K_i)$ is the midpoint of the bid-ask spread.

The VIX is a measure of 30 day implied volatility, how does this relate to our K_{Var}. We can write:

$$VIX = 100\sqrt{K_{Var}}$$

rearrange to get

$$\left(\frac{VIX}{100}\right)^2 = K_{Var}$$

Updating from my previous version of my product, I will now add another comparison later 30-day variance swaps compared to 1 year variance swaps, which I use the VI1X instead of the VIX. This is an area I am going to add to this project in the future deriving a volatility term structure and how this could be useful in modelling future VRP. researching in volatility term structure and modelling and trading behind that.

4 Conclusion

Here, I have derived the replication formula and shown its similarity to the VIX methodology. Now this is an apt place to lead onto our next parts where we investigate how to forecast volatility and then after get onto back testing this variance swap strategy.

5 References

Citation: Demeterfi, K., Derman, E., Kamal, M., Zou, J. (1999). "More Than You Ever Wanted to Know About Volatility Swaps." Goldman Sachs Quantitative Strategies Research Notes (March 1999

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